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Patent Public Search | Text View

United States Patent Application Publication

20250276690

Kind Code

A1

Publication Date

September 04, 2025

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SYSTEM AND METHOD FOR DOWNHILL HEADWAY CONTROL FOR CRUISE CONTROL

Abstract

A cruise control method includes determining that a host vehicle is traveling downhill and determining an axle torque command to maintain a predetermined headway distance from the host vehicle to a lead vehicle while the host vehicle is traveling downhill. The method further includes determining that a brake of the host vehicle is actuated to provide the axle torque command previously determined and comparing the axle torque command with a brake exit threshold to determine whether the axle torque command is greater than the brake exit threshold. Further, the method includes comparing the headway distance between the host vehicle and the lead vehicle with a headway exit threshold to determine whether the headway distance between the host vehicle and the lead vehicle is greater than the headway exit threshold. Also, the method includes commanding the brake of the host vehicle to disengage to terminate braking.

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Family ID: 96738262

Appl. No.: 18/594356

Filed: March 04, 2024

Publication Classification

Int. Cl.: B60W30/16 (20200101); B60W10/18 (20120101); B60W50/00 (20060101)

U.S. Cl.:

CPC B60W30/16 (20130101); B60W10/18 (20130101); B60W50/00 (20130101); B60W2050/0083 (20130101); B60W2300/14 (20130101); B60W2520/403 (20130101); B60W2552/15 (20200201); B60W2554/802 (20200201)

Background/Summary

INTRODUCTION

[0001] The present disclosure relates to adaptive cruise control. Specifically, the present disclosure describes a system and method for downhill headway control using cruise control.

[0002] This introduction generally presents the context of the disclosure. Work of the presently named inventors, to the extent it is described in this introduction, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against this disclosure. When a host vehicle travels downhill, it is challenging to maintain a headway distance between the host vehicle and a lead vehicle due to the use of two actuators (i.e., the propulsion and braking systems) and the gravitational forces due to the road grade. Because two actuators are at play during the downhill movement of the host vehicle, two actuation control commands are necessary to maintain the headway distance between the host vehicle and the lead vehicle. When maintaining this headway distance, the gravitational forces cause the host vehicle **10** to accelerate toward the lead vehicle. If the road is steep enough and long enough, braking may be necessary to maintain the headway distance between the host vehicle and the lead vehicle. Once the desired headway distance is reached, the braking command is terminated, and the host vehicle will once again accelerate toward the lead vehicle. This braking and propulsion cycle, however, causes, among other things, driver discomfort and oscillatory control of the headway distance. It is therefore desirable to develop a method and system to maintain the headway distance from the host vehicle **10** to the lead vehicle that avoids the oscillatory nature of the braking and propulsion cycles.

SUMMARY

[0003] The present disclosure describes a cruise control method. The cruise control method also includes determining that a host vehicle is traveling downhill and determining an axle torque command to maintain a predetermined headway distance from the host vehicle to a lead vehicle while the host vehicle is traveling downhill along a roadway. The lead vehicle is immediately in front of the host vehicle. The axle torque command is a function of a road grade of the roadway. The roadway has a road surface. The road grade is the angle from the road surface to a virtual horizontal line. The method further includes determining that a brake of the host vehicle is actuated to provide the axle torque command previously determined and comparing the axle torque command with a brake exit threshold to determine whether the axle torque command is greater than the brake exit threshold. Further, the method includes, in response to determining that the axle torque command is greater than the brake exit threshold, comparing the headway distance between the host vehicle and the lead vehicle with a headway exit threshold to determine whether the headway distance between the host vehicle and the lead vehicle is greater than the headway exit threshold. Also, the method includes, in response to determining that the predetermined headway distance between the host vehicle and the lead vehicle is greater than the headway exit threshold, commanding the brake of the host vehicle to disengage to terminate braking. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods. The method described in this paragraph improves vehicle technology by maintaining a predetermined distance between a host vehicle and a lead vehicle when the host vehicle is traveling downhill and relying on cruise control, thereby minimizing the risk that the host vehicle will contact the lead vehicle when traveling downhill.

[0004] In some aspects of the present disclosure, the cruise control method may include determining a maximum axle torque command. The maximum torque command is a function of the road grade. Also, the method includes, in response to determining that the axle torque command is

not greater than the brake exit threshold, limiting the axle torque command to the maximum axle torque command. The delta velocity is the difference between the velocity of the lead vehicle and the velocity of the host vehicle. The cruise control method may include comparing the delta velocity with a lead velocity threshold to determine whether the delta velocity is greater than the lead velocity threshold. Further, the cruise control method may include determining that the brake exit threshold is equal to a difference between an initial brake exit threshold and a deceleration offset value in response to determining that the delta velocity is greater than the lead velocity threshold. The cruise control method may include determining that the headway exit threshold is equal to a difference between an initial headway exit threshold and a headway offset value in response to determining that the delta velocity is greater than the lead velocity threshold. The cruise control method may include determining that the brake exit threshold is equal to an initial brake exit threshold in response to determining that the delta velocity is not greater than the lead velocity threshold. The cruise control method may include determining that the headway exit threshold is equal to an initial headway exit threshold in response to determining that the delta velocity is not greater than the lead velocity threshold. The axle torque command is a function of the predetermined headway distance from the host vehicle to the lead vehicle. The vehicle is attached to a trailer. The axle torque command is determined using a following equation:

$$AxCommand=K1*HD.sub.err+K2*\Delta HD.sub.err-mg*\sin(\theta)$$

where: [0005] m =a mass of vehicle plus the mass of trailer; [0006] $K1$ is a first calibration factor; [0007] $K2$ is a second calibration factor; [0008] g is the gravitational constant; [0009] V is the host vehicle speed; [0010] HT is the driver selected headway time; [0011] HD is the predetermined headway distance ($V*HT$) from the host vehicle to the lead vehicle; [0012] L is the measured distance from the host vehicle to the lead vehicle; [0013] $HD.sub.err$ is the headway error, $HD-L$; [0014] θ is the angle from the road surface to the virtual horizontal line; and [0015] $\Delta HD.sub.err$ is a first derivate of the headway error with respect to time.

[0016] The present disclosure also describes a vehicle including sensors, a propulsion system, a braking system, and a controller. The controller is in communication with the sensors, the braking system, and the propulsion system and is programmed to execute the method described above.

[0017] The present disclosure also describes a tangible, non-transitory, machine-readable medium, comprising machine-readable instructions, that when executed by a processor, cause the processor to execute the method described above.

[0018] Further areas of applicability of the present disclosure will become apparent from the detailed description provided below. It should be understood that the detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

[0019] The above features and advantages, and other features and advantages, of the presently disclosed system and method are readily apparent from the detailed description, including the claims, and exemplary embodiments when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

[0021] FIG. 1 is a schematic diagram of a host vehicle including a cruise control system.

[0022] FIG. 2 is a schematic diagram of a host vehicle attached to a trailer, wherein the host vehicle is traveling downhill and is spaced apart from the lead vehicle, and the lead vehicle is also traveling downhill.

[0023] FIG. 3 is a flowchart of a cruise control method for maintaining a headway distance between a host vehicle and a lead vehicle while the host vehicle is traveling downhill.

[0024] FIG. 4 is a flowchart of a method for determining a brake exit threshold and a headway exit threshold used in the method of FIG. 3.

DETAILED DESCRIPTION

[0025] Reference will now be made in detail to several examples of the disclosure that are illustrated in accompanying drawings. Whenever possible, the same or similar reference numerals are used in the drawings and the description to refer to the same or like parts or steps.

[0026] With reference to FIG. 1, a host vehicle **10** generally includes a body **12** and a plurality of wheels **14** coupled to the body **12**. The host vehicle **10** may be an autonomous vehicle. In the depicted embodiment, the host vehicle **10** is depicted in the illustrated embodiment as a sedan, but it should be appreciated that other vehicles including trucks, coupes, sport utility vehicles (SUVs), recreational vehicles (RVs), etc., may also be used.

[0027] The host vehicle **10** further includes one or more sensors **24** coupled to the body **12**. The sensors **24** sense observable conditions of the exterior environment and/or the interior environment of the host vehicle **10**. As non-limiting examples, the sensors **24** may include one or more cameras, one or more light detection and ranging (LIDAR) sensors, one or more radars, one or more Global Positioning System (GPS) transceivers, one or more tilt sensors, one or more inertial measurement units (IMUs), one or more accelerometers, one or more vehicle speed sensors, one or more wheel speed sensors, one or more yaw rate sensors, one or more gyroscopes, one or more proximity sensors, one or more cameras, one or more ultrasonic sensors, one or more thermal imaging sensors, and/or other sensors **24**. Each sensor **24** is configured to generate a signal that is indicative of the sensed observable conditions (i.e., sensor data) of the exterior environment and/or the interior environment of the host vehicle **10**.

[0028] The host vehicle **10** includes a vehicle controller **34** in communication with the sensors **24**. The vehicle controller **34** includes at least one vehicle processor **44** and a vehicle non-transitory computer readable storage device or media **46**. The vehicle processor **44** may be a custom made or commercially available processor, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the vehicle controller **34**, a semiconductor-based microprocessor (in the form of a microchip or chip set), a macroprocessor, a combination thereof, or generally a device for executing instructions. The vehicle computer readable storage device or media **46** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the vehicle processor **44** is powered down. The vehicle computer-readable storage device or media **46** may be implemented using a number of memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or another electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions, used by the vehicle controller **34** in controlling the host vehicle **10**. The vehicle controller **34** of the host vehicle **10** may be programmed to execute the method **100** (FIG. 3) and the method **200** (FIG. 4) as described in detail below.

[0029] The instructions may include one or more separate programs, each of which comprises an ordered listing of executable instructions for implementing logical functions. The instructions, when executed by the vehicle processor **44**, receive and process signals from sensors, perform logic, calculations, methods and/or algorithms for automatically controlling the components of the host vehicle **10**, and generate control signals to automatically control the components of the host vehicle **10** based on the logic, calculations, methods, and/or algorithms. Although a single vehicle controller **34** is shown in FIG. 1, embodiments of the host vehicle **10** may include a plurality of vehicle controllers **34** that communicate over a suitable communication medium or a combination

of communication mediums and that cooperate to process the sensor signals, perform logic, calculations, methods, and/or algorithms, and generate control signals to automatically control features of the host vehicle **10**. The vehicle controller **34** is part of a system **21** for downhill headway control using full speed range adaptive cruise control.

[0030] The host vehicle **10** includes a propulsion system **26** for propelling the host vehicle **10**. The propulsion system **26** is coupled to one or more wheels **14** and may include an internal combustion engine **28** and/or an electric motor **30**. The propulsion system **26** is in communication with the vehicle controller **34**. Accordingly, the vehicle controller **34** may receive information from the propulsion system **26**. Further, the vehicle controller **34** may send commands to the propulsion system **26**.

[0031] In addition, the host vehicle **10** includes a braking system **32** for decelerating the host vehicle **10**. The braking system **32** is coupled to one or more wheels **14** and includes one or more brakes **36**. The braking system **32** is in communication with the vehicle controller **34**. Accordingly, the vehicle controller **34** may receive information from the braking system **32**. Further, the vehicle controller **34** may send commands to the braking system **32**.

[0032] With reference to FIG. 2, when the host vehicle **10** travels downhill, it is desirable to maintain a predetermined headway distance **38** (e.g., three meters) from the lead vehicle **40**. The lead vehicle **40** is directly and immediately in front of the host vehicle **10** and, therefore, no other vehicle is located between the host vehicle **10** and the lead vehicle **40**. A trailer **42** may be attached to the host vehicle **10** while the host vehicle **10** travels downhill. In this case, the host vehicle **10** travels along a roadway **48** having a road surface **50**. The road grade of the roadway **48** is defined as the angle θ from the road surface **58** to a virtual horizontal line **52**.

[0033] FIG. 3 is a method **100** for maintaining the predetermined headway distance **38** between the host vehicle **10** and the lead vehicle **40** while the host vehicle **10** is traveling downhill. The method **100** begins at block **102**. At block **102**, the vehicle controller **34** determines whether the host vehicle **10** is traveling downhill. To do so, the vehicle controller **34** may receive road-grade data from the sensors **24** (e.g., one or more tilt sensors) and/or GPS data from the GPS transceivers. The vehicle controller **34** then uses the road-grade data to determine that the host vehicle **10** is traveling downhill. The road-grade data may include the road grade (i.e., the angle θ from the road surface **58** to a virtual horizontal line **52**) of the roadway **48**. Also, the vehicle controller **34** determines an axle torque command to maintain the predetermined headway distance **38** from the host vehicle **10** to the lead vehicle **40** while the host vehicle **10** is traveling downhill along the roadway **48**. As discussed, the lead vehicle **40** is immediately in front of the host vehicle **10**. The axle torque command is a function of a road grade of the roadway **48**. As discussed above, the road grade is defined as the angle θ from the road surface **58** to a virtual horizontal line **52**. The axle torque command may be a braking command. During a braking command, the vehicle controller **34** commands the braking system **32** to actuate the brakes **36** to decelerate the host vehicle **10**. The vehicle controller **34** may use the following equation to determine the axle torque command:

$$AxCommand = K1 * HD.sub.err + K2 * \Delta HD.sub.err - mg * \sin(\theta)$$

where: [0034] m is the mass of the host vehicle **10** plus the mass of trailer **42**; [0035] $K1$ is a first calibration factor; [0036] $K2$ is a second calibration factor; [0037] g is the gravitational constant; [0038] V is the host vehicle speed; [0039] HT is the driver selected headway time; [0040] HD is the predetermined headway distance ($V * HT$) **38** from the host vehicle to the lead vehicle; [0041] L is the measured distance from the host vehicle to the lead vehicle; [0042] $HD.sub.err$ is the headway error, $HD - L$; [0043] θ is the angle from the road surface **50** to the virtual horizontal line **52**; and [0044] $\Delta HD.sub.err$ is a first derivative of the headway error **38** with respect to time. [0045] The method **100** then proceeds to block **104**. At block **104**, the vehicle controller **34** determines whether one or more brakes **36** of the braking system **32** of the host vehicle are actuated to provide the axle torque command previously determined to maintain the predetermined headway

distance **38** from the host vehicle **10** to the lead vehicle **40**.

[0046] If one or more brakes **36** are not actuated, then the method **100** proceeds to block **104**. At block **104**, the vehicle controller **34** sets the braking actuation flag to be false. Then, the method **100** continues to block **106**. At block **106**, the method **100** restarts. If the one or more brakes **36** are actuated, then the method **100** proceeds to block **108**.

[0047] At block **108**, the vehicle controller **34** determines the maximum axle torque command. The maximum axle torque command is the maximum allowed braking command in terms of acceleration and is a function of the road grade. The maximum axle torque command may be determined by using a look-up table. This look-up table may be developed by testing the host vehicle **10** and has the road grade and the acceleration as the inputs and the maximum axle torque command as the output. Alternatively, the vehicle controller **34** may use the following equation to determine the maximum axle torque command:

$$\text{AxCommandMax} = \max(0, g \cdot \sin(\theta))$$

where: [0048] AxCommandMax is the maximum axle torque command; [0049] g is the gravitational constant; and [0050] θ is the angle from the road surface **50** to the virtual horizontal line **52**, wherein θ is defined as a positive value when traveling downhill and a negative value when traveling uphill, and $g \cdot \sin(\theta)$ is a positive value.

[0051] Then, the method **100** proceeds to block **110**. At block **110**, the vehicle controller **34** determines a brake exit threshold, which is described in detail with respect to FIG. **4**. Then, the method **100** continues to block **112**. At block **112**, the vehicle controller **34** determines the headway exit threshold which is described in detail with respect to FIG. **4**. Next, the method **100** continues to block **114**.

[0052] At block **114**, the controller **34** compares the axle torque command with the brake exit threshold to determine whether the axle torque command is greater than the brake exit threshold. If the axle torque command is greater than the brake exit threshold, then the method **100** proceeds to block **116**. At block **116**, the vehicle controller **34** compares the current headway distance from the host vehicle **10** to the lead vehicle **40** with a headway exit threshold to determine whether the current headway distance from the host vehicle **10** to the lead vehicle **40** is greater than the headway exit threshold. If the current headway distance from the host vehicle **10** to the lead vehicle **40** is greater than the headway exit threshold, then the method **100** proceeds to block **104**. At block **104**, the vehicle controller **34** commands the braking system **32** (and therefore one or more brakes **36**) of the host vehicle **10** to disengage to terminate braking. If the current headway distance from the host vehicle **10** to the lead vehicle **40** is not greater than the headway exit threshold, then the method **100** proceeds to block **118**. At block **118**, the vehicle controller **34** limits the axle torque command to the maximum axle torque command. Therefore, at block **118**, braking is not terminated but it is limited to the maximum axle torque command. After block **118**, the method **100** continues to block **106**.

[0053] FIG. **4** is a flowchart of a method **200** for determining a brake exit threshold and a headway exit threshold used in the method **100** of FIG. **3**. The method **200** begins at block **202**. Then, the method **200** proceeds to block **204**. At block **104**, the vehicle controller **34** determines an initial brake exit threshold. The initial brake exit threshold is determined using a look-up table that is developed by testing the host vehicle **10**. The input of the look-up table is the road grade of the roadway **48**. Therefore, the initial brake exit threshold is a function of the road grade. Then, the method **200** continues to block **206**.

[0054] At block **206**, the vehicle controller **34** determines a deceleration offset value. As discussed below, the deceleration offset value is used if the velocity of the lead vehicle **40** is greater than the velocity of the host vehicle and, therefore, the lead vehicle **40** is moving away from the host vehicle **10**. The deceleration offset value is determined using a look-table that is developed by testing the host vehicle **10**. The input of this look-up table is the road grade of the roadway **48**.

Therefore, the deceleration offset value is a function of the road grade. Then, the method **300** continues to block **208**.

[0055] At block **208**, the vehicle controller **34** determines an initial headway exit threshold. The initial headway exit threshold is determined using a look-up table that is developed by testing the host vehicle **10**. The input of the look-up table is the road grade of the roadway **48**. Therefore, the initial headway exit threshold is a function of the road grade. Then, the method **200** continues to block **210**.

[0056] At block **210**, the vehicle controller **34** determines a headway offset value. The headway offset value is determined using a look-table that is developed by testing the host vehicle **10**. The input of this look-up table is the road grade of the roadway **48**. Therefore, the headway offset value is a function of the road grade. Then, the method **300** continues to block **212**.

[0057] At block **212**, the vehicle controller **34** determines a delta velocity. The delta velocity is the difference between the velocity of the lead vehicle **40** and the velocity of the host vehicle **10**. Then, the vehicle controller **34** compares the delta velocity with a lead velocity threshold to determine whether the delta velocity is greater than the lead velocity threshold. If the delta velocity is greater than the lead velocity threshold, then the method **200** proceeds to block **214**. If the delta velocity is not greater than the lead velocity threshold, then the method **200** proceeds to block **218**.

[0058] At block **214**, the vehicle controller **34** sets the brake exit threshold to be equal to a difference between the initial break exit threshold and the deceleration offset value in response to determining that the delta velocity is greater than the lead velocity threshold. Then, the method **200** continues to block **216**.

[0059] At block **216**, the vehicle controller **34** sets the headway exit threshold to be equal to a difference between an initial headway exit threshold and a headway offset value in response to determining that the delta velocity is greater than the lead velocity threshold. Then, the method **200** continues to block **222**. At block **222**, the method **200** ends.

[0060] At block **218**, the vehicle controller **34** sets the brake exit threshold to be equal to the initial break exit threshold. Then, the method **200** continues to block **220**. At block **220**, the vehicle controller **34** sets the initial headway exit threshold to be equal to the initial headway exit threshold. Then, the method **200** continues to block **222**. At block **22w**, the method **200** ends.

[0061] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the presently disclosed system and method that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and can be desirable for particular applications.

[0062] The drawings are in simplified form and are not to precise scale. For purposes of convenience and clarity only, directional terms such as top, bottom, left, right, up, over, above, below, beneath, rear, and front, may be used with respect to the drawings. These and similar directional terms are not to be construed to limit the scope of the disclosure in any manner.

[0063] Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments can take various and

alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to display details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the presently disclosed system and method. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

[0064] Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by a number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with a number of systems, and that the systems described herein are merely exemplary embodiments of the present disclosure.

[0065] For the sake of brevity, techniques related to signal processing, data fusion, signaling, control, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

[0066] This description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims.

Claims

1. A cruise control method, comprising: determining that a host vehicle is traveling downhill; determining an axle torque command to maintain a predetermined headway distance from the host vehicle to a lead vehicle while the host vehicle is traveling downhill along a roadway, wherein the lead vehicle is immediately in front of the host vehicle, the axle torque command is a function of a road grade of the roadway, the roadway has a road surface, and the road grade is an angle from the road surface to a virtual horizontal line; determining that a brake of the host vehicle is actuated to provide the axle torque command previously determined; comparing the axle torque command with a brake exit threshold to determine whether the axle torque command is greater than the brake exit threshold; in response to determining that the axle torque command is greater than the brake exit threshold, comparing a current headway distance between the host vehicle and the lead vehicle with a headway exit threshold to determine whether the current headway distance between the host vehicle and the lead vehicle is greater than the headway exit threshold; and in response to determining that the predetermined headway distance between the host vehicle and the lead vehicle is greater than the headway exit threshold, commanding the brake of the host vehicle to disengage to terminate braking.

2. The cruise control method of claim 1, further comprising: determining a maximum axle torque command, wherein the maximum axle torque command is a function of the road grade; and in response to determining that the axle torque command is not greater than the brake exit threshold, limiting the axle torque command to the maximum axle torque command.
3. The cruise control method of claim 2, further comprising: determining a velocity of the host vehicle; determining a velocity of the lead vehicle; and determining a delta velocity, wherein the delta velocity is a difference between the velocity of the lead vehicle and the velocity of the host vehicle.
4. The cruise control method of claim 3, further comprising: comparing the delta velocity with a lead velocity threshold to determine whether the delta velocity is greater than the lead velocity threshold.
5. The cruise control method of claim 4, further comprising: setting the brake exit threshold to be equal to a difference between an initial brake exit threshold and a deceleration offset value in response to determining that the delta velocity is greater than the lead velocity threshold.
6. The cruise control method of claim 5, further comprising: setting the headway exit threshold to be equal to a difference between an initial headway exit threshold and a headway offset value in response to determining that the delta velocity is greater than the lead velocity threshold.
7. The cruise control method of claim 4, further comprising: determining that the brake exit threshold is equal to an initial brake exit threshold in response to determining that the delta velocity is not greater than the lead velocity threshold.
8. The cruise control method of claim 7, further comprising: determining that the headway exit threshold is equal to an initial headway exit threshold in response to determining that the delta velocity is not greater than the lead velocity threshold.
9. The cruise control method of claim 1, wherein the axle torque command is a function of the predetermined headway distance from the host vehicle to the lead vehicle.
10. The cruise control method of claim 1, wherein the host vehicle is attached to a trailer, and the axle torque command is determined using a following equation:
$$AxCommand = K1 * HD.sub.err + K2 * \Delta HD.sub.err - mg * \sin(\theta)$$
where: m=a mass of vehicle plus the mass of trailer; K1 is a first calibration factor; K2 is a second calibration factor; g is a gravitational constant; V is the host vehicle speed; HT is the driver selected headway time; HD is the predetermined headway distance (V*HT) from the host vehicle to the lead vehicle; L is the measured distance from the host vehicle to the lead vehicle; HD.sub.err is the headway error, HD-L; θ is the angle from the road surface to the virtual horizontal line; and $\Delta HD.sub.err$ is a first derivate of the headway error with respect to time.
11. A vehicle, comprising: a plurality of sensors; a propulsion system; a braking system including a brake; a controller in communication with the plurality of sensors, the propulsion system, and the braking system, wherein the controller is programmed to: determine that a host vehicle is traveling downhill; determine an axle torque command to maintain a predetermined headway distance from the host vehicle to a lead vehicle while the host vehicle is traveling downhill along a roadway, wherein the lead vehicle is immediately in front of the host vehicle, the axle torque command is a function of a road grade of the roadway, the roadway has a road surface, and the road grade is an angle from the road surface to a virtual horizontal line; determine that the brake of the host vehicle is actuated to provide the axle torque command previously determined; compare the axle torque command with a brake exit threshold to determine whether the axle torque command is greater than the brake exit threshold; in response to determining that the axle torque command is greater than the brake exit threshold, compare a current headway distance between the host vehicle and the lead vehicle with a headway exit threshold to determine whether the current headway distance between the host vehicle and the lead vehicle is greater than the headway exit threshold; and in response to determining that the predetermined headway distance between the host vehicle and the lead vehicle is greater than the headway exit threshold, command the brake of the host vehicle to disengage to

terminate braking.

12. The vehicle of claim 11, wherein the controller is programmed to: determine a maximum axle torque command, wherein the maximum axle torque command is a function of the road grade; and in response to determining that the axle torque command is not greater than the brake exit threshold, limit the axle torque command to the maximum axle torque command.

13. The vehicle of claim 12, wherein the controller is programmed to: determine a velocity of the host vehicle; determine a velocity of the lead vehicle; and determine a delta velocity, wherein the delta velocity is a difference between the velocity of the lead vehicle and the velocity of the host vehicle.

14. The vehicle of claim 13, wherein the controller is programmed to: compare the delta velocity with a lead velocity threshold to determine whether the delta velocity is greater than the lead velocity threshold.

15. The vehicle of claim 14, wherein the controller is programmed to: sets the brake exit threshold to be equal to a difference between an initial brake exit threshold and a deceleration offset value in response to determining that the delta velocity is greater than the lead velocity threshold.

16. The vehicle of claim 15, wherein the controller is programmed to: sets the headway exit threshold to be equal to a difference between an initial headway exit threshold and a headway offset value in response to determining that the delta velocity is greater than the lead velocity threshold.

17. The vehicle of claim 14, wherein the controller is programmed to: determine that the brake exit threshold is equal to an initial brake exit threshold in response to determining that the delta velocity is not greater than the lead velocity threshold.

18. The vehicle of claim 17, wherein the controller is programmed to: determine that the headway exit threshold is equal to an initial headway exit threshold in response to determining that the delta velocity is not greater than the lead velocity threshold.

19. The vehicle of claim 11, wherein the axle torque command is a function of the predetermined headway distance from the host vehicle to the lead vehicle.

20. The vehicle of claim 11, wherein the vehicle is attached to a trailer, and the axle torque command is determined using a following equation:

$AxCommand = K1 * HD.sub.err + K2 * \Delta HD.sub.err - mg * \sin(\theta)$ where: m=a mass of host vehicle plus the mass of trailer; K1 is a first calibration factor; K2 is a second calibration factor; g is a gravitational constant; V is the host vehicle speed; HT is the driver selected headway time; HD is the predetermined headway distance (V*HT) from the host vehicle to the lead vehicle; L is the measured distance from the host vehicle to the lead vehicle; HD.sub.err is the headway error, HD-L; θ is the angle from the road surface to the virtual horizontal line; and $\Delta HD.sub.err$ is a first derivate of the headway error with respect to time.
